

lot Smart City

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Introduction

- Cities are hubs of innovation and service provision, drawing migrants from rural areas. With over 55 percent of the global population residing in cities (projected to reach 68 percent by 2050), urbanization is a key driver of economic development. However, rapid growth poses challenges like infrastructure demands, unemployment, and environmental impact.
- Urban sprawl contributes to increased energy consumption, air pollution, and reduced open space. Managing these challenges requires smart city concepts, integrating advanced technologies to enhance information flow, resident quality of life, and sustainable economic development. The fourth industrial revolution, marked by the Internet of Things (IoT), further transforms smart cities.

Smart Cities

- Smart city concepts, shaped by technological advances, began with the digitization of cities in the 1950s, leading to the rise of terms like "computable cities" and "cyber cities" in the '90s. The term "smart city" gained popularity with the spread of the Internet and information technologies.
- The integration of physical and virtual environments became crucial for smart city development, attracting attention from researchers, policymakers, and urban planners. Despite various terms used, "smart city" has become the most widely adopted.
- While lacking a universal definition, smart cities are often described through six dimensions, extending beyond technology to include governance, economy, living, people, mobility, and environment. Technology, especially advancements like cyber-physical systems and the Internet of Things, is integral to achieving smart city goals.

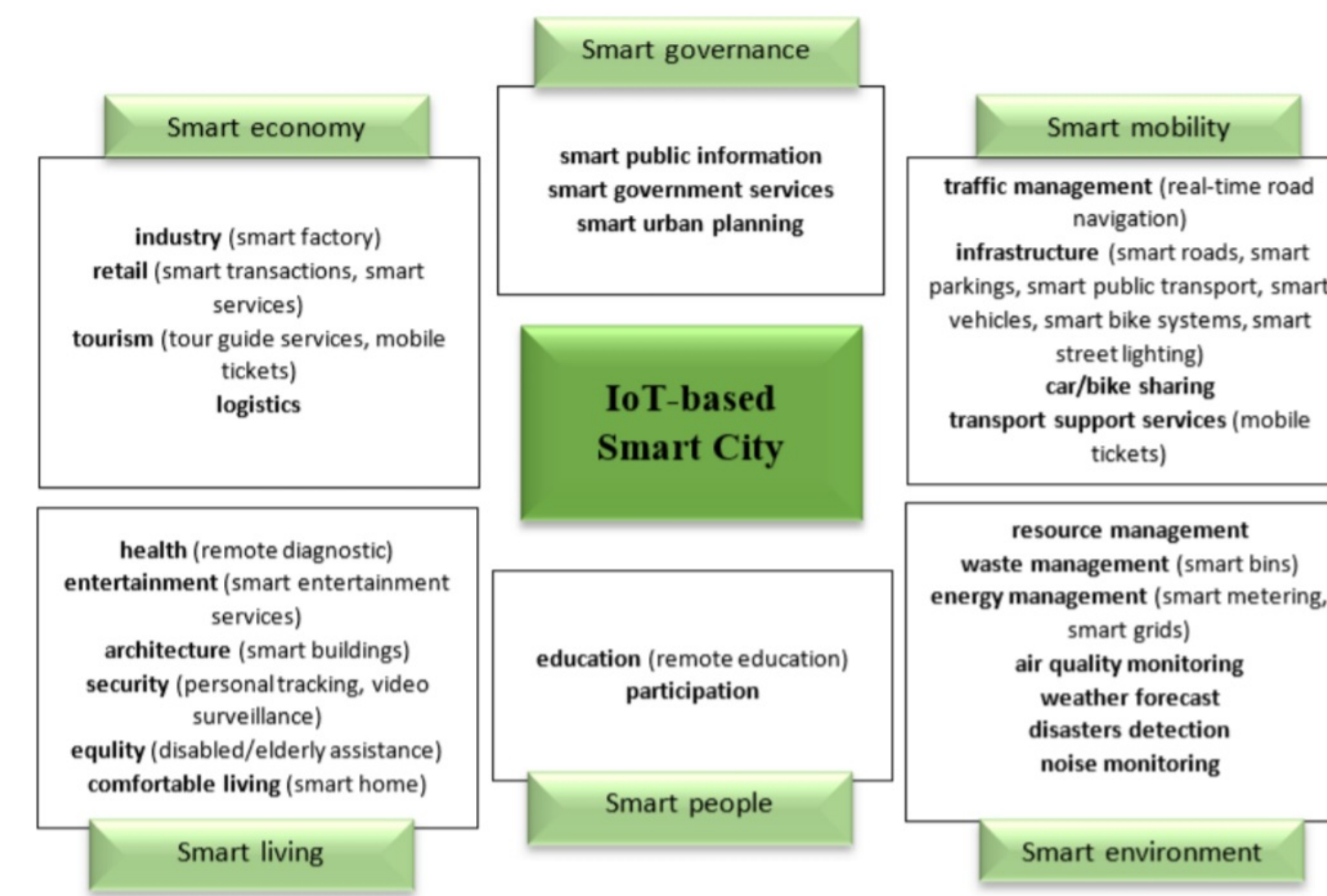


Figure 1: Applications of IoT in Smart Cities

Internet of Things

- Coined by Kevin Ashton in 1999, the term "Internet of Things" (IoT) envisions a world where physical objects connect to the Internet, enabling real-time two-way information exchange.
- IoT forms an integrated network facilitating communication between objects and users through embedded sensors, processors, and various communication solutions. Definitions of IoT vary, ranging from a technical focus to considering interactions between technical and social elements.
- Incorporating technologies like cloud computing, fog computing, edge computing, RFID, WSN, machine-to-machine communication, and IPv6, IoT's architecture comprises five layers: perception (device), network (transmission), middleware, application, and business layers.
- IoT's potential applications in daily life are extensive, spanning from smart health and education to smart buildings, industry, agriculture, and more. In smart cities, IoT enhances services like education, healthcare, public safety, waste management, environmental monitoring, and transportation.

Challenges Of IoT

- Privacy and Security Challenges: Information: Use of identification technologies (RFID, 2D barcode) poses risks of private information leakage. Objects: Large number of objects at different locations increases the risk of unauthorized interference. Network: Transmission of vast data over wireless networks may result in overload and unauthorized access.
- Data Encryption: Ensuring secure encryption of data.
- Object Management: Switching on/off, changing target connection networks, detecting failures, updating software.
- Identity Management: Challenges arise from numerous objects connected to the network, creating issues with nomenclature and identification.
- Integration: Connecting multiple devices with different parameters and specifications can hinder achieving system-wide interoperability.
- Optimization of Energy Consumption: Increasing numbers of smart devices and data transfer speeds are associated with higher energy consumption.
- Implementation Challenges in Smart Cities: Influenced not only by technological issues but also by social, economic, and political aspects. Factors determining smart city development include the digital skills of urban residents, openness to change, effectiveness of city authorities, and investment in research and development.

Conclusion

- The vision of smart cities driven by the Internet of Things (IoT) is becoming a reality, promising benefits such as streamlined processes, enhanced decision-making, time savings, and cost minimization across various human activities. However, implementing the IoT paradigm in urban environments presents engineering and scientific challenges.
- While technology is crucial for smart city development, authors emphasize that technology alone does not define a city's intelligence; it should serve as a tool to address real urban challenges.

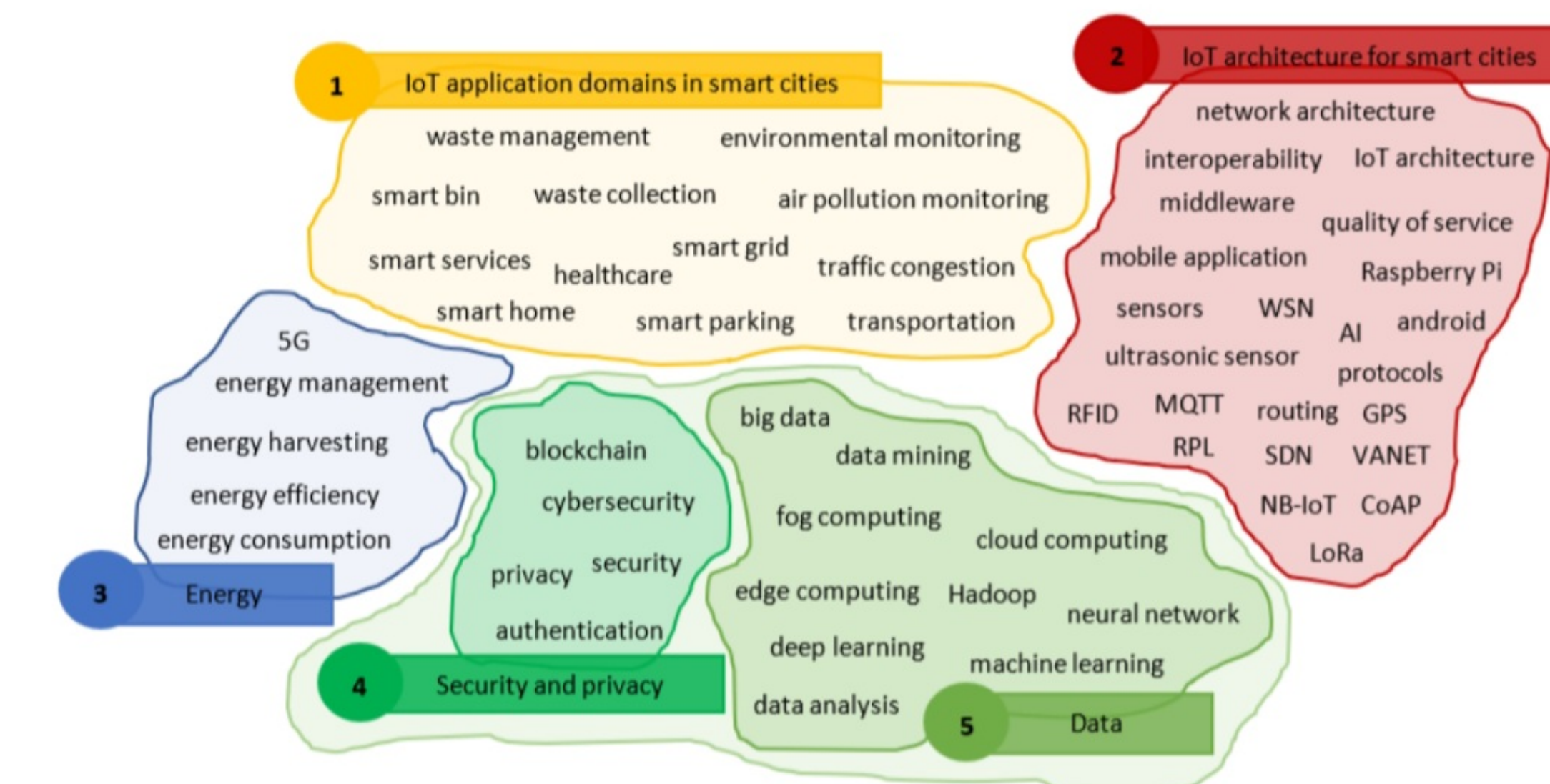


Figure 2: Subareas of "IoT-based" smart city research